Fiscal Year:	FY 2007	**Task Last Updated:**	FY 11/09/2009
PI Name:	Walker, Thad G Ph.D.		
Project Title:	All-optical high density atom sources		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	53706	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2001	**End Date:**	04/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	4	**No. of Master' Degrees:**	2
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 4/30/2007 per PI (7/2006)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and uncover mysteries of the universe obscured by Earth's gravity		

Our approach to producing high density, high phase space density atoms is to laser cool atoms confined in a holographically produced far-off-resonance lattice. This Holographic Atom Trap (HAT) is produced by imaging a hologram onto a cloud of laser-cooled atoms. The HAT produces strongly confining forces on atoms in a 10-100 micron volume. Once in the HAT, the atoms will be further cooled using lasers to temperatures of a few micro-Kelvin.
The ultimate densities and phase space densities that can be reached in the HAT depend on the collisional properties of the trap and heating due to photon scattering. Both effects are minimized in the HAT by optical pumping into dark states. We will investigate the possibility of direct laser cooling to quantum degeneracy in the HAT.

We seek to use the HAT as a source for studying quantum transport through atom waveguides. Our waveguide will be

Task Description:	formed by a laser beam with a hole down the middle. Characteristic steps in the transmission of the guide as a function of energy are quantized in units of the DeBroglie wavelength. The small energy scales involved mean that gravity is a substantial complication, and later manifestations of the experiment will be substantially improved in a microgravity environment. We seek to develop new high density, high phase space density sources of cold atoms for scientific exploitation of microgravity environments. Such sources potentially have inherent advantages as compared to currently used technology, namely speed, atom efficiency, and power consumption. We propose to develop and study the properties of these sources, and to use them to address a fundamental physics problem ideally suited to microgravity: quantization of atom conductance through an atom waveguide. Development of high-density atom sources by laser cooling will have advantages over current technology in terms of speed, efficiency, and power consumption. These new sources may be of considerable interest to a wide variety of experiments using ultracold atoms on the ground and in space. This project will focus on the properties of Rydberg states of ultracold atoms inside a Holo- graphic Atom Trap (HAT). Having acheived atom densities in excess of 10e15 cm-3 , and cloud sizes of 8 µm or less using evaporative cooling and adiabatic compression, the project will have a unique and promising laboratory for studying a variety of Rydberg phenomena. We will study the so-called blockade effect, where the presence of even a single Rydberg atom in the trap is predicted to preclude excitation of additional Rydberg atoms using the same laser fields. The blockade has tremendous promise for a variety of novel phenomena such as fast quantum state detection and transmission, single photon and single atom sources, and as a entanglement mechanism for neutral atom quantum computation. Narrow-band two-photon excitation of Rydberg states in high density atom clouds is predicted to be strongly affected by the blockade collisional effect in which the presence of a single excited Rydberg state can frequency shift neighboring atoms out of resonance with the excitation light. A striking prediction is that mesoscopic clouds of thousands of atoms will behave as a single quantum entity, producing single Rydberg atoms on demand. The single atom character of the process will be clearly demonstrated using photon antibunching measurements. Furthermore, collective enhancement of the two-photon Rabi frequency will demonstrate the coherent nature of the blockade effect. The fundamental nature of the blockade will be investigated. In particular novel “Forster zero” states are predicted to circumvent the blockade effect for certain types of Rydberg states. Using electric fields, dual atom resonances can be tuned to enhance the Forster mechanism and perhaps destroy the degeneracies responsible for the zeros. Taking advantage of the highly anisotropic trapping potential in the HAT, high quality dipole blockade is predicted to be possible even when the Rydberg atoms are fully hybridized into giant permanent dipoles.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research is important for developing new means of producing cold atoms, and for learning how to use these cold atoms to make novel quantum systems that may be of interest for a variety of applications, including quantum information processing and clocks. Students trained using these studies are well positioned to meet the nation's needs for scientific expertise.
Task Progress:	In PHYSICAL REVIEW A 66, 06540(2002), we discussed the application of dipole blockade techniques for the preparation of single-atom and single-photon sources. A deterministic protocol was given for loading a single atom in an optical trap, as well as ejecting a controlled number of atoms in a desired direction. A single-photon source with an optically controlled beamlike emission pattern was described. In OPTICS LETTERS 28, 149 (2003), we demonstrated the production of high-density cold 87 Rb samples (2.3X10^14 atom/cm^3) in a simple optical lattice formed with YAG light that is diffracted from a holographic phase plate. A loading protocol was described that results in 10,000 atoms per 10 µmX10 µm X100 µm unit cell of the lattice site. Rapid free evaporation led to a temperature of 50 µK and phase space densities of 1/150 within 50 ms. The resulting small, high-density atomic clouds are very attractive for a number of exper iments, including ultracold Rydberg atom physics. In J. Phys. B: At. Mol. Opt. Phys. 38 (2005) S309–S319, we showed that, even though long-range Rydberg–Rydberg interactions can inhibit multiple excitations of atoms under the appropriate conditions (blockade), in many situations the dipole-dipole interaction vanishes so that care is required to realize full dipole blockade in micron-sized atom samples. In PHYSICAL REVIEW A 71, 0214R (2005), we demonstrated the production of micron-sized high-density atom clouds of interest for mesoscopic quantum information processing. We evaporate atoms from 6 µK, 3X10^14 atoms / cm^3 samples contained in a highly anisotropic optical lattice formed by interfering diffracted beams from a holographic phase plate. After evaporating to 1 µK by lowering the con?ning potential, in less than a second the atom density reduced to 8X10^13 cm^-3 at a phase space density approaching unity. Adiabatic recompression of the atoms then increased the density to levels in excess of 1X 10^15 cm^-3. The resulting clouds were typically 8 µm in the longest dimension. Such samples are small enough to enable mesoscopic quantum manipulation using the Rydberg blockade and have the high densities required to investigate collision phenomena. In PHYSICAL REVIEW A 77, 052712 (2008), population dynamics in weakly excited clouds of ultracold 87Rb Rydberg atoms were studied by means of trap loss, fluorescence detection, and state-dependent stimulated emission. Rydberg atoms were excited to various nl Rydberg states via continuous two-photon excitation from a magneto-optical trap. A stimulated emission probe laser was then used to bring the Rydberg atoms down to the 6P3/2 state, allowing state-dependent detection of the Rydberg atoms. Measurements of trap loss and fluorescent emission reveal information about the evolution of the Rydberg populations. In particular, population in the initial Rydberg state quickly transfers to other Rydberg states by a noncollisional mechanism, likely superradiant emission. The trap-loss measurements are consistent with black-body ionization as the dominant loss mechanism.
Bibliography Type:	Description: (Last Updated: 11/10/2009)

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